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Airport (20_582) Homework 3

1 Problem

The following data are given for passenger travel demand over a five-year period.

Table 1: Passenger Travel Demand Data

Fare per Passenger-Mile, p	Travel Demand, q (10^6 passenger-miles)	Year, t
0.30	639	1386
0.30	698	1387
0.30	762	1388
0.30	828	1389
0.30	904	1390

1. Using regression analysis, estimate the travel demand volume for the year 1395.
2. Suppose that, due to a new regulation introduced in Farvardin 1391, the fare per passenger-mile decreases to $p = 0.24$. As a result, the travel demand in 1391 increases to 1,221,000,000 passenger-miles.

Answer the following questions:

- (a) Calculate the demand elasticity.
- (b) Determine a linear demand function of the form:

$$q = f(p)$$

for the year 1391.

- (c) Determine the maximum profit that the airport can obtain in 1391, assuming the total cost function is:

$$TC = 100,000 + 80,000q$$

- (d) Determine a demand function of the form:

$$q = f(p, t)$$

for the travel demand volume.

- (e) What additional information would be necessary for designing the airport transportation system corresponding to the airports discussed in part (a)?

1.1 solution to part a

To estimate the travel demand for year 1395, a linear regression model is fitted in the form:

$$q = a + b(t - 80)$$

where:

- q = travel demand (10^6 passenger-miles)
- t = year

For simplicity, define:

$$x = t - 80$$

Thus, the data become:

$$(6, 639), (7, 698), (8, 762), (9, 828), (10, 904)$$

The required summations are:

$$\sum x = 6 + 7 + 8 + 9 + 10 = 40$$

$$\sum q = 639 + 698 + 762 + 828 + 904 = 3831$$

$$\sum x^2 = 36 + 49 + 64 + 81 + 100 = 330$$

$$\sum xq = 6(639) + 7(698) + 8(762) + 9(828) + 10(904) = 31308$$

and

$$n = 5$$

The slope of the regression line is:

$$b = \frac{n \sum xq - (\sum x)(\sum q)}{n \sum x^2 - (\sum x)^2}$$
$$b = \frac{5(31308) - 40(3831)}{5(330) - 40^2} = \frac{156540 - 153240}{1650 - 1600} = \frac{3300}{50} = 66$$

The intercept is:

$$a = \frac{\sum q - b \sum x}{n}$$
$$a = \frac{3831 - 66(40)}{5} = \frac{3831 - 2640}{5} = 238.2$$

Therefore, the regression equation becomes:

$$\boxed{q = 238.2 + 66(t - 80)}$$

For year 1395:

$$t - 80 = 15$$

Thus:

$$q_{1395} = 238.2 + 66(15)$$

$$q_{1395} = 238.2 + 990$$

$$\boxed{q_{1395} = 1228.2}$$

Therefore, the estimated travel demand for year 1395 is:

$$\boxed{1228.2 \times 10^6 \text{ passenger-miles}}$$

1.2 solution to part b

1.2.1 Demand Elasticity

From part (a), the estimated demand for year 1391 at the original fare $p = 0.30$ is:

$$q_1 = 238.2 + 66(11) = 964.2$$

After the fare reduction:

$$p_2 = 0.24$$

and the new demand becomes:

$$q_2 = 1221$$

Using the arc elasticity formula:

$$E = \frac{\Delta q}{\Delta p} \cdot \frac{\bar{p}}{\bar{q}}$$

First compute the changes and averages:

$$\Delta q = 1221 - 964.2 = 256.8$$

$$\Delta p = 0.24 - 0.30 = -0.06$$

$$\bar{p} = \frac{0.30 + 0.24}{2} = 0.27$$

$$\bar{q} = \frac{964.2 + 1221}{2} = 1092.6$$

Therefore:

$$E = \frac{256.8}{-0.06} \cdot \frac{0.27}{1092.6}$$

$$\boxed{E \approx -1.06}$$

The negative sign indicates that demand increases when price decreases.

1.2.2 Linear Demand Function for Year 1391

Using the two points:

$$(p, q) = (0.30, 964.2)$$

and

$$(p, q) = (0.24, 1221)$$

Assume a linear demand function:

$$q = mp + c$$

The slope is:

$$m = \frac{1221 - 964.2}{0.24 - 0.30} = \frac{256.8}{-0.06} = -4280$$

Thus:

$$q = -4280p + c$$

Using point $(0.30, 964.2)$:

$$964.2 = -4280(0.30) + c$$

$$964.2 = -1284 + c$$

$$c = 2248.2$$

Therefore, the demand function becomes:

$$q = 2248.2 - 4280p$$

1.2.3 Maximum Profit in 1391

From the demand function:

$$q = 2248.2 - 4280p$$

Express price in terms of demand:

$$p(q) = \frac{2248.2 - q}{4280}$$

Revenue is:

$$R = 10^6 pq$$

Thus:

$$R = 10^6 \left(\frac{2248.2 - q}{4280} \right) q$$

The total cost function is given as:

$$TC = 100000 + 80000q$$

Profit is:

$$\Pi(q) = R - TC$$

Therefore:

$$\Pi(q) = 10^6 \left(\frac{2248.2 - q}{4280} \right) q - 100000 - 80000q$$

Simplifying:

$$\Pi(q) = -233.64486q^2 + 445280.3738q - 100000$$

To maximize profit:

$$\frac{d\Pi}{dq} = 0$$

Hence:

$$-467.28972q + 445280.3738 = 0$$

Solving:

$$q^* = 952.9$$

Substituting into the demand equation:

$$p^* = \frac{2248.2 - 952.9}{4280}$$

$$p^* \approx 0.3026$$

Finally:

$$\Pi_{\max} = 10^6 \left(\frac{2248.2 - 952.9}{4280} \right) (952.9) - 100000 - 80000(952.9)$$

$$\Pi_{\max} \approx 2.12 \times 10^8$$

1.2.4 Demand Function of the Form $q = f(p, t)$

Assume a linear demand model:

$$q = A + B(t - 80) + Cp$$

From part (a), the time trend coefficient is:

$$B = 66$$

Thus:

$$q = A + 66(t - 80) + Cp$$

Using the two observations for year 1391:

$$964.2 = A + 66(11) + 0.30C$$

$$1221 = A + 66(11) + 0.24C$$

Since:

$$66(11) = 726$$

the equations become:

$$964.2 = A + 726 + 0.30C$$

$$1221 = A + 726 + 0.24C$$

Subtracting the equations:

$$256.8 = -0.06C$$

$$C = -4280$$

Substituting back:

$$964.2 = A + 726 - 1284$$

$$A = 1522.2$$

Therefore, the final demand model becomes:

$$q = 1522.2 + 66(t - 80) - 4280p$$

1.2.5 Additional Information Required for Airport Transportation System Design

To design the airport transportation system, annual passenger demand alone is not sufficient. Additional information is required, including:

- Peak-hour passenger demand
- Seasonal demand variations
- Origin-destination passenger distribution
- Passenger type (domestic, international, transfer)
- Aircraft size and fleet mix
- Passenger load factors
- Desired level of service

These data are necessary for proper design of airport terminals, runways, parking areas, and transportation facilities.